Compression and Cost in High-Resolution X-Ray CT

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Advances in X-ray detector hardware have led to a substantial increase in matrix sizes, which now commonly reach 4k × 4k, with the resulting canonical X-ray Computed Tomography (CT) reconstruction volumes exceeding 100 GB for 16-bit voxels. This places considerable strain on imaging facilities and their users, driven by the storage costs of high-resolution X-ray imaging.

The compression ratios (CR), compression and decompression times, and quality loss are determined for in-memory compression tests across a large number of lossless and lossy compression schemes, including JPEG-LS and the recent JPEG XL standard. Relevant subsets of the tests are performed in TIFF, DICOM, and (OME-)Zarr containers in order to determine the most viable archival formats and their compatibility with prevalent image processing software packages. Quality loss is quantified using common full-reference metrics including SSIM[1], MS-SSIM[2], and PSNR.

We find that lossless schemes yield a maximal CR of 1.75:1, corresponding to a 43% space saving (SS). The JPEG-LS and JPEG XL codecs achieve CRs of 16.3:1 (93.9% SS) and 45.6:1 (97.8% SS) respectively at imperceptible quality loss (PSNR $\approx$ 60 dB full scale, 3D SSIM $\approx$ 0.999) on a representative 16-bit CT volume. Alongside JPEG 2000, these schemes in combination with a DICOM container[3] constitute the only lossy options widely supported among the surveyed packages. Compression times remain compatible with high-throughput acquisition, while decompression is fast enough that access times improve over the uncompressed, I/O-bound case.

To illustrate the economic implications, we estimate storage savings for the NaturArv3D collaborative project[1] between the DTU 3D Imaging Centre and the Natural History Museum of Denmark. Under realistic operating conditions, the expected savings amount to approximately 1 PB of storage annually relative to the existing data pipeline, which corresponds to an estimated annual cost reduction of 3 million DKK. This underscores that the adoption of lossy compression schemes is decisive for the economic viability of high-throughput X-ray imaging at scale.

References

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